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SMART INDIA
HACKATHON
2026



ग्रामीण विकास मंत्रालय
MINISTRY OF
RURAL DEVELOPMENT

Integrated GIS-based Digital Public Infrastructure for Land Governance

Transparent | Secure | Accessible | Inclusive

*“A Smarter Tomorrow
for Every Piece of Land”*

Team name : Last not Least

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Institute : Tripur Institute of Technology

Problem Statement ID : SIH26014



The Problem We Are Solving



Fragmented
land records



Time-consuming
manual processes



No real-time
visibility into land
status



Multiple departments
and databases



Citizens face
disputes, fraud
and poor access



Lack of real-time
spatial information



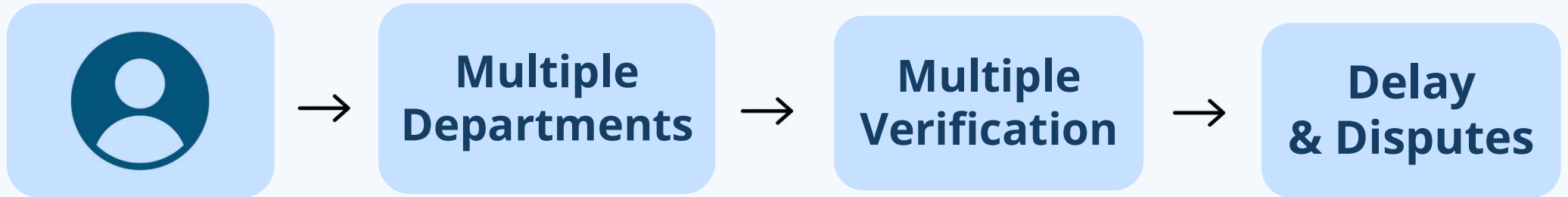
Inconsistent and
outdated records



Outdated
Lands

***“ Land Information exists, but it is not sufficiently
connected, accessible & actionable ”***

Current System & Its Gaps



Existing Approach

Paper/Manual records

Separate databases

Multiple Verification

Limited GIS integration

Physical visit

Limited transparency



Problems

Slow & error-prone

Data fragmentation

Time Consuming

Poor spatial understanding

Inconvenient

Disputes/fraud risk



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Our Proposed Solution



“ One integrated platform connections citizen, land records, GIS data & authorities.”

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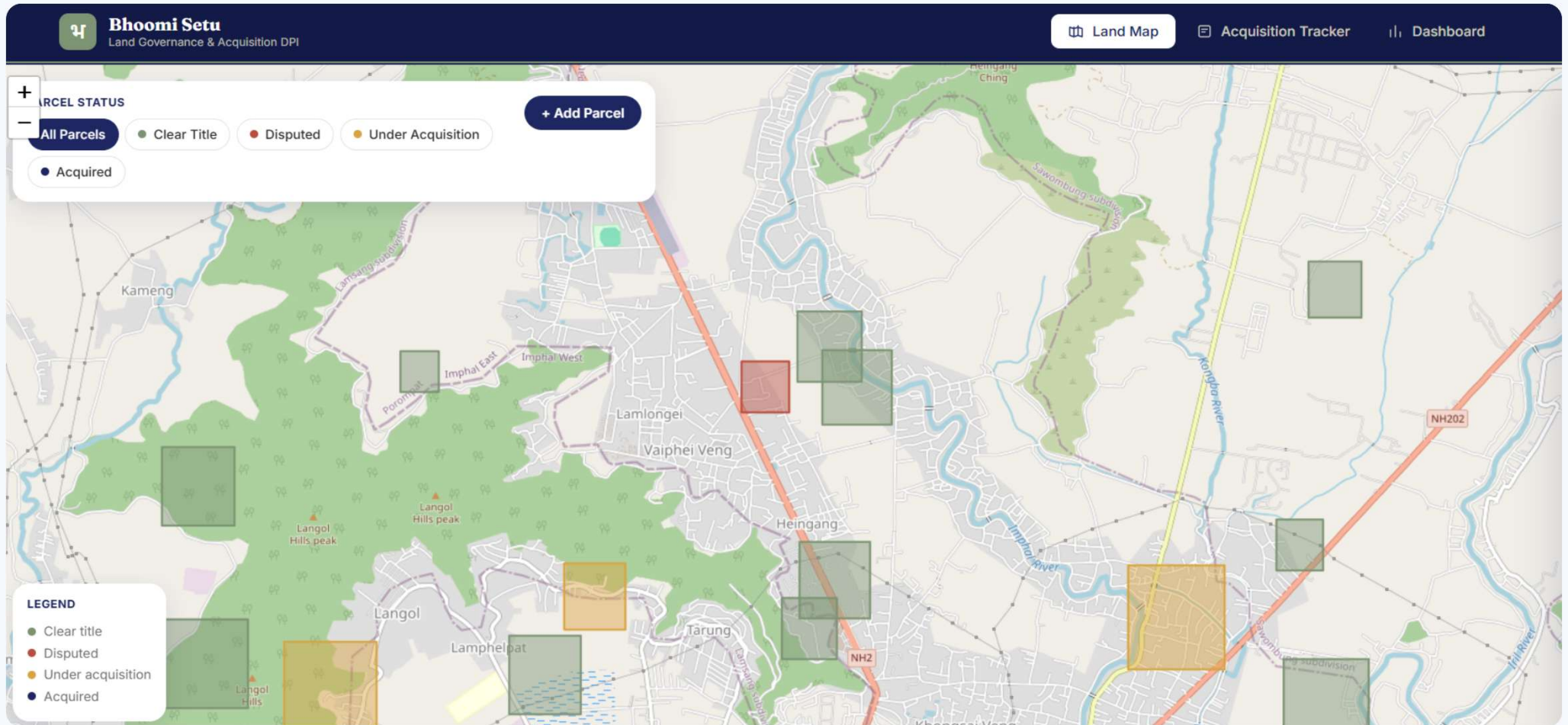
How the Platform Works



From search to settlement – All in one platform



GIS-Based Land Map

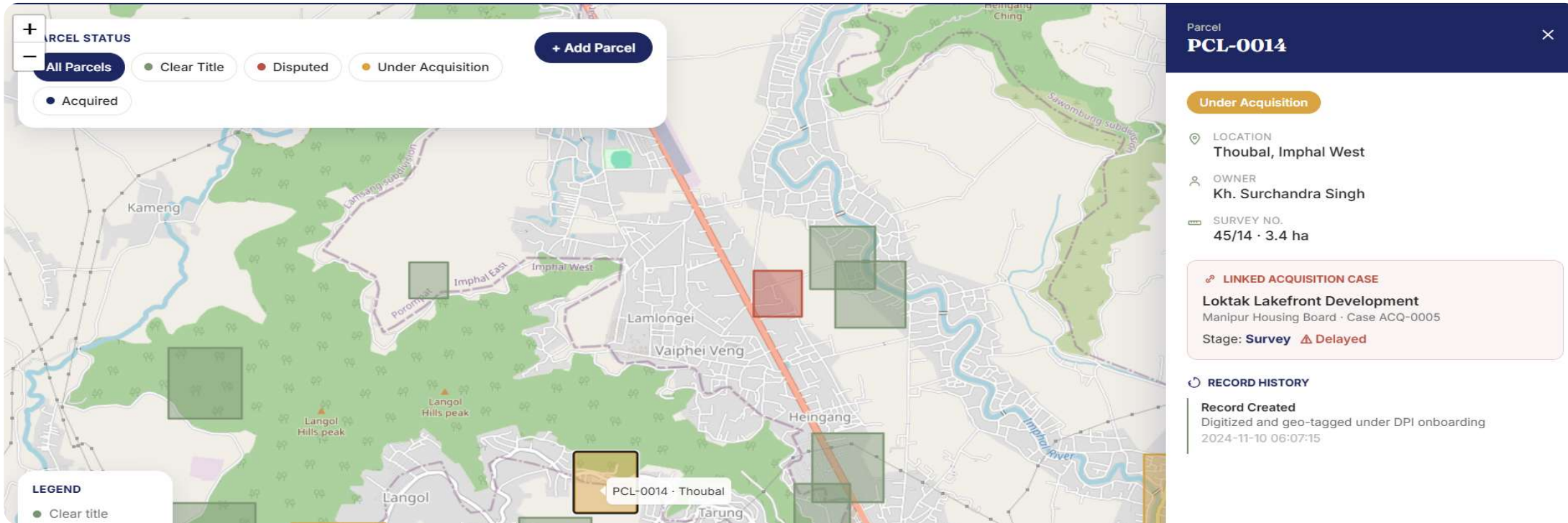


Every land parcel becomes a digitally identifiable spatial asset



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GIS-Based Land Map



Secure Land Marketplace



Sell Land

- ✓ List your property
- ✓ Add land details
- ✓ Upload documents
- ✓ Get verified
- ✓ Reach genuine buyers

List Property



Buy Land

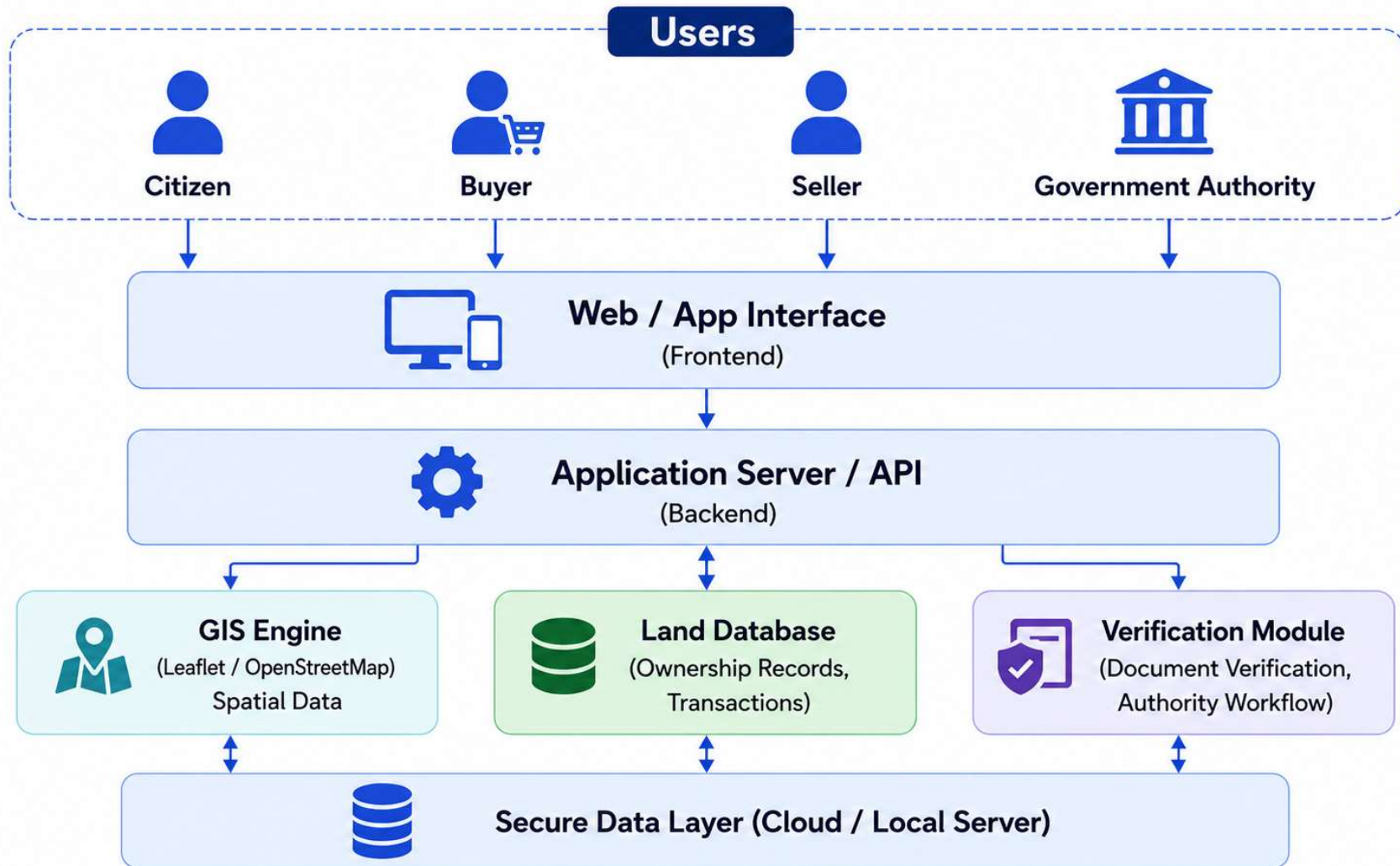
- ↑ Search on GIS map
- ↑ View verified details
- ↑ Check documents
- ↑ Submit purchase request
- ↑ Track status

Search Property

A transparent & secure marketplace for land transaction



System Architecture



Security, Transparency & Trust



11 Expected Impact



Processing Time



Transparency



Fraud Risk



Citizen Accessibility



Manual Paperwork



Data Accuracy

Beneficiaries



Citizens



Land Owners



Buyers



Sellers



Government
Authorities



Survey/Revenue
Departments



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Working Prototype

Coming soon.....

13 Technical Approach

Technologies Used



Frontend

HTML, CSS, JavaScript / React



Backend

Node.js, Express.js REST APIs



Database

PostgreSQL / MongoDB



GIS

Leaflet / OpenStreetMap



Authentication

Secure role-based
user authentication



Document Verification

Digital document upload and
verification workflow

Methodology



- Collect and digitize existing land records



- Link every land parcel with a unique Plot ID



- Integrate GIS coordinates with ownership
and property records



- Verify ownership through authority /
document verification



- Enable secure buying, selling and
transaction tracking



- Maintain audit trails for transparency

14 Feasibility & Viability



Analysis of Feasibility

- Uses widely available and scalable web/GIS technologies
- Can integrate with existing government land databases
- Modular architecture allows gradual implementation
- GIS-based system can support both urban and rural areas



Potential Challenges

- Incomplete or outdated existing land records
- Integration with multiple government databases
- Data privacy and cybersecurity requirements
- Verification of historical ownership documents
- Differences in land-record formats between states



Strategies to Overcome Challenges

- Digitization and standardization of legacy records
- API-based integration with government systems
- Role-based access and encrypted data storage
- Multi-level authority verification
- Audit logs for every modification and transaction
- Phased deployment starting with pilot districts



15 Impacts & Benefits



Potential Impact

- Faster and more transparent land-related services
- Reduced land disputes and fraudulent transactions
- Improved accessibility of land information for citizens
- Better coordination between departments and authorities
- Digital transformation of traditional land governance

Benefits



Social

- Easier access to land records
- Greater citizen transparency
- Reduced dependency on physical visits



Economic

- Faster property transactions
- Reduced administrative costs
- Increased confidence in land markets



Environmental

- Reduced paper-based documentation
- Better GIS-based land-use planning
- Support for sustainable development planning



Government

- Centralized and accurate land information
- Better monitoring and decision-making
- Reduced manual paperwork and processing time



Business Model



Government (B2G)

- Annual SaaS license
- GIS digitization services
- Implementation & support



Businesses (B2B)

- Subscription access
- API services
- Verified land data
- Analytics & insights



Citizens (B2C)

- Free basic services
- Paid verification reports
- Document services
- Marketplace access



Transaction Revenue

- Small service fee on successful deals
- Premium land verification
- Custom reports

Future Scope



Integration with government land databases



Aadhaar/Identity verification (as per regulations)



Digital signatures



Blockchain-based transaction records



Satellite imagery integration



AI-based document analysis



Mobile application



Automated dispute detection



Integration with registration departments



Multi-state deployment



Environmental & land-use planning



Advanced analytics and dashboards

18 Research & Reference



Government & Official Sources

- Digital India Land Records Modernization Programme (DILRMP)
- Department of Land Resources, Ministry of Rural Development
- Bhunaksha / NIC GIS-based cadastral mapping systems
- Digital India initiative
- SVAMITVA Scheme – Ministry of Panchayati Raj



Technology References

- OpenStreetMap & Leaflet GIS documentation
- PostgreSQL / PostGIS spatial database documentation
- REST API architecture and secure web application practices
- Research on GIS-based land administration and cadastral systems



Alternative References & Standards

- UN-GGIM & FIG Guidelines: Global Land Tool Network (GLTN) framework and International Federation of Surveyors (FIG) standards for cadastral domain models (ISO 19152 LADM).
- Open Geospatial Consortium (OGC): OGC Web Map Service (WMS), Web Feature Service (WFS), and GeoJSON specifications for spatial data interoperability.





One Unified DPI

Land records + acquisition,
one shared database



Live GIS Mapping

Real parcels, real status,
on an interactive map



Auto-Synced Records

Acquisition updates flow
into land records
instantly



Govt-Ready Model

Built like public DPI —
UPI & DigiLocker pattern

A working prototype today — a scalable land governance backbone for tomorrow



Citizens



Government



Banks / NBFCs



Project Authorities

Thank You

Question ?



**Smart India
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